Computational Methods - Problem Set I

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(a) Relationship between announced prize and revenue

We first get our database from ⟨https://loterias.caixa.gov.br/Paginas/Mega-Sena.aspx⟩ in section "Downloads". To estimate the relationship between the announced prize and the revenue collected, we estimate the following log-log regression:

$$\log(R_j) = \alpha + \beta \log(Y_j) + \varepsilon_j$$

where R_j is the total revenue collected in draw j and Y_j is the announced prize.

β corresponds to the elasticity of revenue with respect to the announced prize.

$$\hat{\beta} \approx 40\%$$

	Dependent variable:
	log(Revenue)
log(Prize)	0.402***
	(0.006)
Constant	10.938***
	(0.097)
Observations	1,900
R^2	0.713
Adjusted R^2	0.713
Residual Std. Error	0.312 (df = 1898)
F Statistic	4,726.452*** (df = 1; 1898)
Note:	*p<0.1; **p<0.05; ***p<0.01

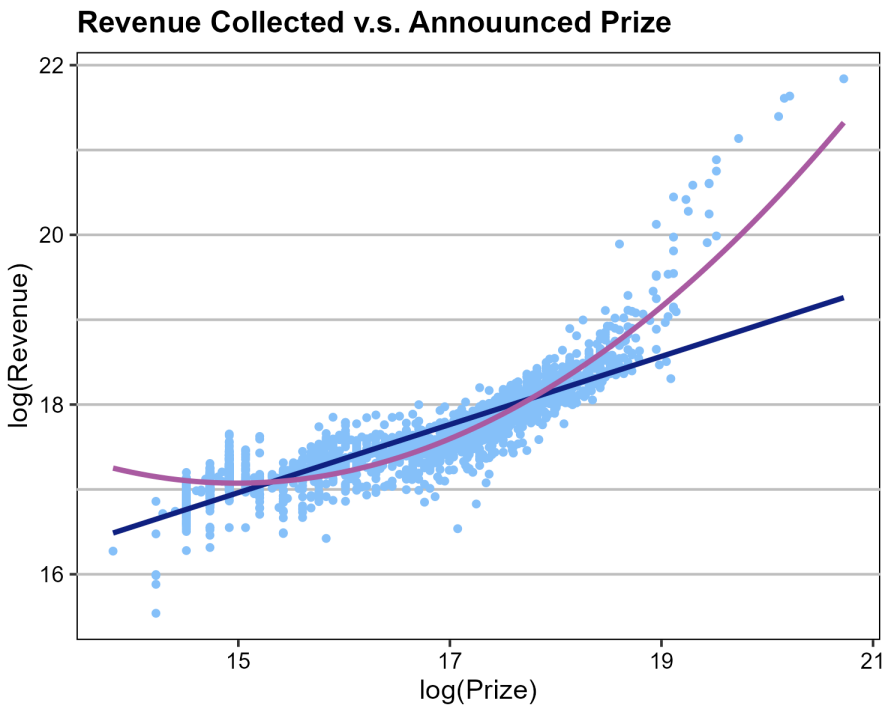

Figure 1: Relationship between announced prize and revenue

(b) Number of individuals participating

To estimate the number of individuals participating in each draw, we assume that every participant buys the cheapest possible ticket, which consists of choosing six numbers.

Therefore, the number of equivalent single tickets in draw j can be computed as

$$N_j = \frac{R_j}{p_j}$$

where p_j is the price of a single Mega-Sena ticket in draw j.

The dataset provided does not contain the ticket price for each draw. To obtain this information, I collected the historical ticket prices from ⟨https://graficos.poder360.com.br/PEXGE/1/⟩.

Finally, the number of equivalent single tickets was computed by dividing the total revenue of each draw by the corresponding ticket price.

(c) Expected prize

For each draw, the expected prize of a single ticket is given by the weighted average of the prizes for 6, 5, and 4 correct numbers. Formally:

$$\mathbb{E}[Y_j] = \frac{G_{6j}}{N_j} Y_{6j} + \frac{G_{5j}}{N_j} Y_{5j} + \frac{G_{4j}}{N_j} Y_{4j}$$

where G_{bj} is the number of individuals that got b numbers correct and Y_{bj} is the prize value received by each individual with b correct numbers, with $b \in \{4, 5, 6\}$.

(d) Return of buying a ticket

The return of buying a ticket is defined as the ratio between the expected prize and the ticket price. Formally, the return in draw j is given by:

$$\text{Return}_j = \frac{\mathbb{E}[Y_j]}{p_j}$$

This measure represents the expected monetary payoff per unit of money spent on a ticket.

(e) Expected return conditional on the announced prize

We are interested in estimating the conditional expectation

$$\mathbb{E}\left[\frac{\mathbb{E}[Y_j]}{p_j} \mid P_j\right]$$

where P_j denotes the announced prize in draw j.

To estimate this relationship, we group the draws into bins according to the value of the announced prize. For each bin we compute the average return and plot it for each prize bin.

Additionally, we estimate a parametric specification by fitting a quadratic regression

$$\frac{\mathbb{E}[Y_j]}{p_j} = \alpha + \beta_1 P_j + \beta_2 P_j^2 + \varepsilon_j$$

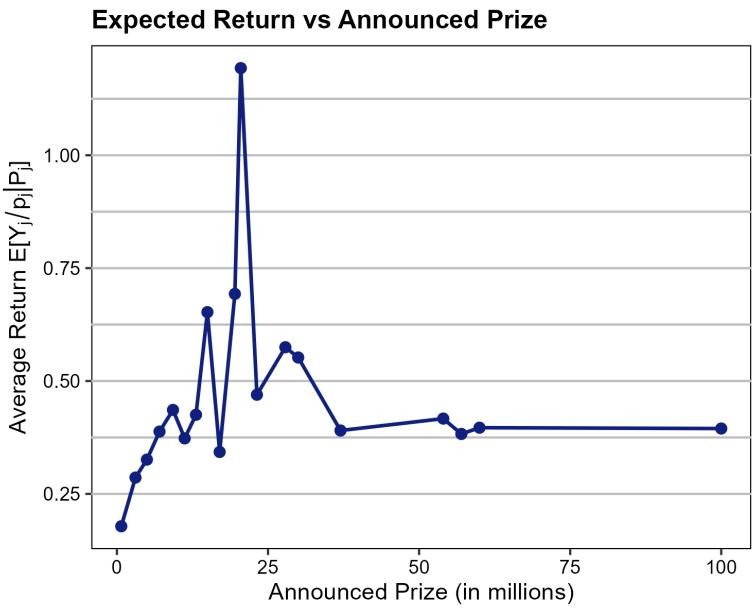

Figure 2: Expected Return with Bins

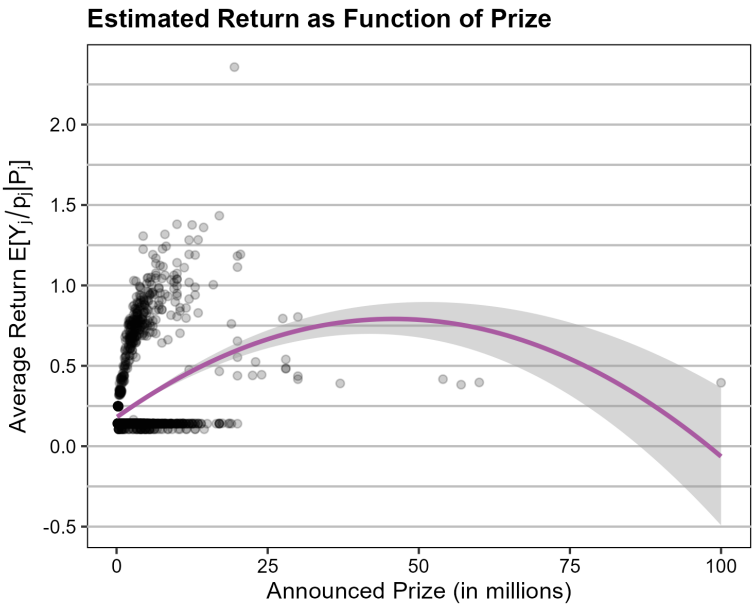

Figure 3: Expected Return in Parametric Form

Both approaches show that the expected return generally increases with the announced prize for small and medium jackpot values. When the announced prize is very small, the expected return is relatively low, reflecting the fact that the total prize pool is limited while the probability of winning remains extremely small.

As the announced prize increases, the expected return rises because larger jackpots increase the potential payoff from correctly guessing the six numbers. However, the relationship becomes much flatter for larger jackpots. This occurs because higher prizes attract more participants, increasing the number of tickets played and therefore the probability that the prize will be shared among multiple winners.

There is only one spike where the estimated return slightly exceeds one. This result is likely driven by a small number of draws in that range, producing a noisy estimate rather than a systematic positive expected value.